

Curriculum Vitae

Siddharth Setlur

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Personal Information

Name:	Siddharth Setlur	Phone:	+44 0735 609 38 90
Date of Birth:	18. Dec 2000	Languages:	English (first), German (fluent)
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Education

Sept 2024 –	PhD at CDT in Algebra, Geometry and Quantum Fields, University of Edinburgh Supervised by Dr. Darrick Lee & Dr. Sjoerd Beentjes. Fully funded by the UKRI ESPRC.
Sept 2022 – August 2024	Master of Science (with distinction) at ETH Zürich in Mathematics MSc Thesis: <i>Optimizing Functions Based on Signed Barcodes</i> Supervised by Dr. Sara Kališnik & Dr. Steve Oudot.
Sept 2019 – Aug 2022	Bachelor of Science at ETH Zürich in Mathematics BSc Thesis: <i>Different Theoretical Perspectives on Persistent Homology</i> Supervised by Dr. Sara Kališnik.

Research Interests

Broadly speaking, I am interested in applying what are generally considered to be pure mathematical tools to machine learning and biomedicine. For example, much of my past work has been in topological data analysis, where I worked on developing new loss functions based on topological invariants in the hopes of creating ML models that optimize for certain geometric structures. This is important in fields such as drug discovery, where one seeks to find molecules with certain shapes.

Currently, I primarily focus on new category theoretic frameworks for causal inference. The goal is two fold: firstly, we hope to better model higher order causal interactions with this framework. Secondly, we hope that this new framework can be used to extend recent work that uses causal inference as a theoretical foundation for mechanistic interpretability.

Publications

† = first or joint first authorship.

- {Luis Scoccola, Siddharth Setlur}[†], David Loiseaux, Mathieu Carrière, and Steve Oudot. **Differentiability and Convergence of Filtration Learning with Multiparameter Persistence.** International Conference on Machine Learning (ICML) 2024.

Work Experience

Oct 2025 – March 2026	Research Assistant, Quantum Software Lab, University of Edinburgh Contributed mathematical expertise to the QSL project entitled <i>Quantum Machine Learning for Genomics: Assessing Real-World Potential and Constraints</i> .
Jan 2024 – April 2024	Teaching Assistant, Department of Mathematics, University of Edinburgh Tutored a weekly course on titled <i>Advanced Research Computing</i> , which aimed to introduce machine learning methods to pure mathematicians. In particular, I helped design the assignments and exercises for the topological deep learning section.
Sept 2023 – Dec 2023	Teaching Assistant, Department of Mathematics, ETH Zürich Teaching a twice-weekly exercise class for first-year Math and Physics students taking Dr. Alessio Figalli’s Single Variable Analysis course.
Oct 2022 – Feb 2023	Research Intern, Inria Saclay (France) Supervised by Dr. Steve Oudot. Research in theoretical foundations of topological data analysis, namely the differentiability of new invariants for multiparameter persistence. Funded by National Institute for Research in Digital Science and Technology
June 2021 – Sep 2022	Library Assistant, Mathematics Library, ETH Zürich Responsibilities included helping select new books for the collection, opening and closing the library once a week, processing book returns, answering customer queries, and conducting inventory.

Workshops and Conferences Attended

Jan 2026	Foundations of causal inference , Isaac Newton Institute, Cambridge University, UK.
July 2025	SIAM Conference on Applied Algebraic Geometry (AG25) , University of Wisconsin-Madison, U.S.
November 2024	Twentieth Meeting of the Applied Algebra and Geometry Research Network , Oxford University
August 2024	Spires 2024 , Oxford University Poster Session for <i>Differentiability and Convergence of Filtration Learning with Multiparameter Persistence</i> (joint work with Luis Scoccola, David Loiseaux, Mathieu Carrière, and Steve Oudot).
July 2024	The Forty-first International Conference on Machine Learning , Vienna Poster Session for <i>Differentiability and Convergence of Filtration Learning with Multiparameter Persistence</i> (joint work with Luis Scoccola, David Loiseaux, Mathieu Carrière, and Steve Oudot).

August 2023	BIREP Summer School on persistence modules—The Interplay of Representation Theory and Topological Data Analysis Contributed Talk: <i>Signed Barcodes for Multiparameter Persistence</i> .
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Travel Funding Awarded

August 2025	The Geometric Realization of AATRN (Applied Algebraic Topology Research Network) , Chicago, U.S. Awarded USD 1000 towards travel and lodging. Declined due to time constraints.
August 2024	Spires 2024 , Oxford University Awarded a travel fund of GBP 200 and accommodation covered.
August 2023	BIREP Summer School on persistence modules—The Interplay of Representation Theory and Topological Data Analysis Travel expenses, accommodation, and food covered.

Service & Volunteering

Feb 2024 –	Organizer, AGATE Seminar Jointly organized a weekly seminar focused on applications of algebra, geometry, and topology to biology.
October 2024 – Dec 2024	Organizer, BioTop Reading Group Organized a hybrid seminar at the University of Edinburgh focusing on applications of topology, geometry, and algebra to biology.
Feb. 2022 – July 2022	Organizer, Zurich Undergraduate Colloquium in Math and Physics Organized a weekly colloquium where undergraduates were given the opportunity to present talks about their semester/bachelor/master theses.